He also shed light on application-aware storage of data: "Sometimes, data should be completely separate from the application; but in some cases, the application needs to be aware of the data, for example, when it comes to performance issues."

Security comes first

If cloud storage can be accessed anywhere, there should be a focus on security. Storage vendors are adopting important security technologies to protect data from security risks. This includes ransomware protection, and 'air-gapping' of secondary storage from primary storage to prevent attacks on backup and archived data. This is effectively making data storage technology in some cases the last line of defence against attacks.

McDonald said that through geolocation based access controls, we give everyone (good or bad) on the Internet an excuse to go around blocking data as and when they see fit. "We need to be really careful when we start talking about securing data, particularly by geolocation; there are some big ethical questions here. It is one of the things that bothers me enormously," he said.

There is also a concern about where the data is located. "If I have data processing taking place on the network, then the concept of data becomes a bit fuzzy. There are concerns about privacy and the kinds of things we will be able to do in the network to data, without anybody really knowing about it. That's a serious problem as well," he added.

He believes that HTTP/3, a protocol that Google brought to the ITF (International Transport Forum), will develop further and make its way into the network file system (NFS), just as it reached server message block (SMB). These two major file protocols can provide true end-to-end security. "Data privacy is one of the things. The other is standards," he added.

Computational storage

Recollecting the five storage technologies, McDonald said that computational storage is about to transform in terms of focusing on applications and in the way applications access data. He added that computational storage, along with the emerging pattern of data centres and cloud becoming highly distributed, is what we need to watch out for. "Instead of asking for small parts of the data set, we may actually ask the devices themselves to be smart about the data, and tell us what they know about the data they have collected," he explained.

The user may have storage devices that are able to process the data, in order to collect metadata. Then it will be possible to work with the storage device instead of asking every object that's got a metadata tag. "We may be working towards actually asking the device, how many red objects it has got? And if your answer is none, there is no data transfer. It's a simple Knack type message that we can send to a device. And I think we may see a lot more of that in future. We are going to see a DPU (data processing unit) in smart network cards that can transform the data. We can actually process data on the network rather than having to move it to a CPU to get processed or to the GPU to get further smashed and computed on," said McDonald.

Software defined storage

The services related to storage are defined by a software stack that usually runs on commodity hardware. The vendor's hardware can be optimised to run its software stacks or it may have some hardware base that affects storage performance.

Reisner said that with Linbit's focus on software-defined storage and open source, clients are driving the company to adapt its software to the existing high-end hardware. Though its storage devices provide an extremely high number of IOPS (input/output operations per second), there is some concern about how to ensure the software stack can actually take advantage and exploit their functionality and performance attributes. "We have single PCI devices doing like 1.6 million IOPS, and a server can have eight of these. This was a real challenge for me," he added.

The future may see more software stacks available from popular storage brands.

Standardisations and SNIA (Storage Networking Industry Association)

Many of the storage related standardisation problems faced earlier by users continue to-date and have not been fixed. McDonald said, "We may have14 standards but may need another one when we are trying to standardise stuff."

Storage is all about bytes. "I would like to see more of Kubernetes in storage technology. We need to get the right levels of abstraction and the right types of interfaces to storage. Standards bodies like SNIA have a huge role to play and are absolutely essential in this environment," he added.

The storage, management and protection of data will continue to be critical in the future. Businesses will collect data at an ever-increasing rate, and need new ways of keeping it accessible and secure. Storage decisions will always be made based on an organisation's need to grow its operations and generate profit. END

The article has been compiled from a panel discussion held at SODACON 2021.

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